Alien Tiles Problem

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1 CSP Formulation and Solving Approach

The problem was encoded as a Constraint Optimisation Problem in MiniZinc. The decision variables clicks$[i, j] \in \{0, \ldots, c-1\}$ denote how many times each cell is clicked; the domain is restricted to $c-1$ per cell without loss of generality, since clicking a cell c times is congruent to not clicking it at all. For every cell (r, s), a modular linear constraint enforces that the cumulative effect of all row r and column s clicks — corrected for the double count on the intersection cell (r, s) itself — equals the goal value $\mod c$, yielding a system of n^2 simultaneous congruences. The objective function minimises $\sum_{i,j}$ clicks$[i, j]$, directly encoding the "fewest moves" criterion central to the original CSPLib formulation. The model is solved via the Gecode finite-domain constraint solver, invoked programmatically through MiniZinc Python, which performs constraint propagation combined with branch-and-bound search to certify optimality of the returned solution. A notable property, established analytically and verified empirically through exhaustive enumeration on small instances, is that the click operation is not always surjective over $Z_c^{n\times n}$: reachability of an arbitrary goal configuration depends on $\gcd(c, n+1)$, meaning that a non-trivial subset of goal boards is provably unreachable from the all-zero start. Rather than a limitation of the encoding, this is a feature of the symbolic formulation: infeasibility (UNSAT) is derived directly and certifiably from the constraints themselves, providing an explicit, verifiable explanation for why no solution exists.

2 Video Link

3 Running Instructions

4 AI Usage